TRAN HOANG PHUONG LINH
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Data-driven analyst with strong experience in **Python, NLP, machine learning, and reproducible data workflows**. Skilled in transforming large, messy datasets into actionable insights, building analytical tools, and communicating results to technical and non-technical audiences. Experienced in automation, text analytics, forecasting, and end-to-end data pipelines. Passionate about applying quantitative methods to solve economic, financial, and operational problems.

TECHNICAL SKILLS

- **Programming:** Python (pandas, NumPy, scikit-learn, matplotlib, NLP libraries), SQL, MATLAB, Stata, R
- **Data & Analytics:** Data cleaning, feature engineering, time-series analysis, text processing, forecasting
- **Machine Learning:** Tree-based models, regularized models (Lasso, Ridge), CNNs, classification & prediction
- **NLP:** Topic modelling, readability metrics, sentiment analysis, text classification
- **Tools:** Git, Excel (advanced), Power BI (beginner-intermediate), HTML (basic)
- **Workflow:** Reproducible pipelines, automation, documentation**.**
- **Methods:** Econometrics (panel data, IV, nonlinear models), robust inference, causal analysis

SELECTED PROJECTS

Applied Machine Learning: Kickstarter Outcome Prediction

- Built an end-to-end ML pipeline using regularized models, decision trees, and CNNs.
- Engineered features, evaluated multi-class and binary classification models, and optimized performance.
- Delivered a reproducible workflow with clear metrics, visualizations, and documentation.

Central Bank Communication in Times of Crisis (2020–2024)

- Constructed a multi-country panel dataset of central bank statements using Python and web-scraping.
- Extracted linguistic features (readability, abstractness, informativeness, disunity) using NLP tools.
- Modelled relationships between communication complexity and macroeconomic indicators.
- Built visualizations and automated workflows to support analysis.

Complexity in FOMC Communication: Drivers, Differences, and Market Reactions

- Analyzed 7,000+ Federal Reserve policymaker speeches using Python-based NLP and econometric frameworks.
- Extracted linguistic complexity features and linked them to market conditions.
- Identified systematic communication differences across policymakers.
- Evaluated market reactions, showing short-run volatility spikes and longer-run dampening effects.
- Built reproducible pipelines for feature extraction, visualization, and model comparison.

Deliberation and Policy Outcomes: Evidence from the FOMC

- Processed and analyzed FOMC meeting transcripts (1976–2018) using Python-based NLP.
- Developed quantitative measures of policy stance, sentiment, and collaboration.
- Integrated text-derived indicators into an econometric model of committee deliberation.
- Presented findings to diverse audiences and iterated analyses based on feedback.

Online Data Competitions (Kaggle, Analytics Vidhya)

- Built predictive models for retail sales, HR analytics, and language scoring tasks.
- Applied feature engineering, cross-validation, and model comparison techniques

EXPERIENCE

Teaching Assistant — McGill University (2021–2025)

Courses: Econometrics, Quantitative Methods, Applied Cross-Sectional Methods, Economic Statistics

- Delivered tutorials and supported instruction for classes of 20–400 students.
- Designed assessments, ensured consistent grading, and communicated complex concepts clearly.
- Managed deadlines and coordination across multiple courses.

Research Assistant — University of Queensland (2018)

- Cleaned, validated, and analyzed survey and administrative datasets.
- Identified data quality issues and ensured reproducibility of research outputs.
- Prepared summary tables, charts, and documentation for macroeconomic research.

ML Reproducibility Challenge — McGill University (2021)

- Reproduced results from a published ML paper by implementing models and verifying outputs.
- Documented methodology, code, and evaluation metrics in a structured report.

EDUCATION

- **Ph.D. Economics,** McGill University, Canada — *Completed June 2026*
- **B.A. (Honours) Economics & Econometrics,** University of Queensland, Australia — *2019*

ADDITIONAL INFORMATION

- **Languages:** English (fluent), Vietnamese (native), French (beginner)
- **Other Projects:** Regulatory text complexity analysis; causal inference under sparsity SNIV
- **Interests:** NLP, automation, forecasting, data visualization, workflow design